MA1513 Exam (Examplify)

AY2023/2024 Semester I

MCQ/FIB (15 questions; 3 marks each)

1. A 3 by 4 homogeneous linear system has general solution $\begin{cases} x_1 & = & s-2t \\ x_2 & = & t-2s \\ x_3 & = & s \\ x_4 & = & t \end{cases}$. The reduced row echelon form of the augmented matrix of the system is given by $\left(\begin{array}{cccc|c} 1 & 0 & a & b & 0 \\ 0 & 1 & c & d & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$ where a = ___, b = ___, c = ___, d = ___ (FIB)

2. Suppose A = (c_1 c_2 c_3) is a 3x3 singular matrix, where c_1 , c_2 , c_3 are the three columns of A. Which of the following statements are always true? (Pick all correct options)
   a. c_3 is a linear combination of c_2 and c_3.
   b. { c_1 , c_2 , c_3 } forms a basis for $\mathbf{R}^3$.
   c. There is a non-pivot column in any row echelon form of A.
   d. The nullity of A is not 0.
   e. The linear system Ax = b has infinitely many solutions for any 3-vector b.

3. Which of the following have the same value as the 3x3 determinant $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ d & e & f \end{vmatrix}$ for any values of a, b, c, d, e and f? (Pick all correct options)
   a. $\begin{vmatrix} b & c \\ e & f \end{vmatrix} + \begin{vmatrix} a & c \\ d & f \end{vmatrix} + \begin{vmatrix} a & b \\ d & e \end{vmatrix}$
   b. $a\begin{vmatrix} 1 & 1 \\ e & f \end{vmatrix} - b\begin{vmatrix} 1 & 1 \\ d & f \end{vmatrix} + c\begin{vmatrix} 1 & 1 \\ d & e \end{vmatrix}$
   c. $d\begin{vmatrix} 1 & 1 \\ b & c \end{vmatrix} - e\begin{vmatrix} 1 & 1 \\ a & c \end{vmatrix} + f\begin{vmatrix} 1 & 1 \\ a & b \end{vmatrix}$
   d. $\begin{vmatrix} a & b \\ d & e \end{vmatrix} + \begin{vmatrix} b & c \\ e & f \end{vmatrix} + \begin{vmatrix} c & a \\ f & d \end{vmatrix}$
   e. $\begin{vmatrix} b & c \\ e & f \end{vmatrix} - a\begin{vmatrix} 1 & 1 \\ e & f \end{vmatrix} + d\begin{vmatrix} 1 & 1 \\ b & c \end{vmatrix}$

4. The vector (3,6,9) belongs to the row space of which of the following matrices? (Pick all correct answers.)
   a. $\begin{pmatrix} 0 & 0 & 3 \\ 0 & 6 & 0 \\ 9 & 0 & 0 \end{pmatrix}$
   b. $\begin{pmatrix} 3 & 3 & 3 \\ 6 & 6 & 6 \\ 9 & 9 & 9 \end{pmatrix}$
   c. $\begin{pmatrix} 3 & 6 & 0 \\ 0 & 6 & 9 \end{pmatrix}$
   d. $\begin{pmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$
   e. $\begin{pmatrix} 3 & 3 & 3 \\ 0 & 3 & 3 \\ 0 & 0 & 3 \\ 6 & 3 & 0 \end{pmatrix}$

5. Which of the following sets of vectors are bases for the subspace span{(1,2,1,1), (1,0,2,-1), (1,4,0,3), (-1,0,-2,1)} of $\mathbf{R}^4$? (Pick all correct answers)
   a. {(1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0), (0, 0, 0, 1)}
   b. {(1,2,1,1), (1,0,2,-1), (1,4,0,3)}
   c. {(1,0,2,-1), (1,4,0,3)}
   d. {(1,2,1,1), (-1,0,-2,1)}
   e. {(1, 0,2,0), (0, 1, -1, -1)}

6. Which of the following vectors will have the projection p on the plane with equation x – 2y + z = 0 where p = (-1,1,3). (Pick all correct answers)
   a. (1,-2,1)
   b. (3,0,1)
   c. (0,-1,4)
   d. (-2,3,2)
   e. (1,0,-1)
7. Suppose the regular x, y and z-axes in the XYZ space is rotated $45°$ clockwise about the z-axis (when view from top). The coordinate vector of $(3\sqrt{2}, \sqrt{2}, 3)$ relative to the new axes is given by (a, b, c) where a = ___, b = ___, c = ___ (FIB)
   (Hint: The answers are whole numbers with no square root involved)

8. Suppose we are given the reduced row echelon form R of an unknown matrix A. Which of the following conditions allow us to determine the column space of A?
   a. R is full rank.
   b. R does not have any zero row.
   c. R has no non-pivot columns.
   d. Number of rows and columns of R are the same.
   e. It is not possible to determine the column space of A without knowing A.

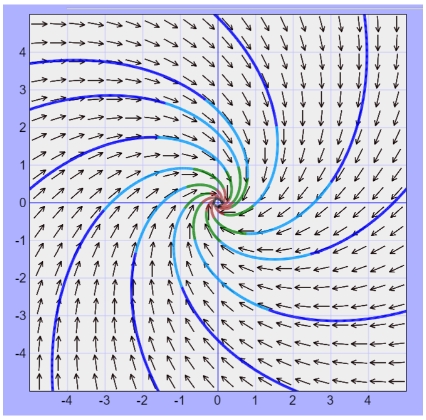
9. (Linear transformation) Let T: $\mathbf{R}^3 \to \mathbf{R}^3$ be a linear transformation with standard matrix A. Suppose the first two columns of A are $\begin{pmatrix}1\\1\\0\end{pmatrix}$ and $\begin{pmatrix}0\\1\\1\end{pmatrix}$. Knowing the image of which of the following vectors will allow us to completely determine T? (Pick all correct answers)
   a. $\begin{pmatrix}1\\1\\0\end{pmatrix}$
   b. $\begin{pmatrix}0\\1\\1\end{pmatrix}$
   c. $\begin{pmatrix}1\\2\\1\end{pmatrix}$
   d. $\begin{pmatrix}1\\0\\0\end{pmatrix}$
   e. Any vector except the linear combinations of $\begin{pmatrix}1\\1\\0\end{pmatrix}$ and $\begin{pmatrix}0\\1\\1\end{pmatrix}$.

10. Given A is a 4x4 matrix and u, v, w, x are linearly independent vectors in R^4. Suppose Au = v, Av = u, Aw = 0 and Ax = x.
Which of the following statements are correct? (Pick all correct answers)
    a. A has 3 distinct eigenvalues
    b. A has only 2 linearly independent eigenvectors
    c. A has an eigenspace with dimension 2
    d. A is not diagonalisable
    e. A is singular

11. In a certain town, 25% of the married men get divorced each year and 20% of the single men get married each year. There are 6000 married men and 3000 single men. Assuming that the total population of men remains constant over the years, there will be M married men and S single men in the long run, where M = ___, S = ___ (FIB).

12. The general solution of a 3 by 3 SDE is given by
$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = c_1 \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} e^{-7t} + c_2 \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} e^{3t} + c_3 \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} e^{t}$. If an initial condition is given by $\begin{pmatrix} y_1(0) \\ y_2(0) \\ y_3(0) \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}$,
then c_1 = ___, c_2 = ___, c_3 = ___ (FIB)

13. The phase portrait of a 2 by 2 SDE y' = Ay is given below.



Which of the following is the underlying matrix A of the SDE?
    a. $\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$
    b. $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$
    c. $\begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix}$
    d. $\begin{pmatrix} -1 & -1 \\ 1 & -1 \end{pmatrix}$
    e. None of the above

14. The 3rd order ODE given by 2y'''+6y'' – 4y' -2y = 0 can be converted to a 3 by 3 SDE given by
$\begin{pmatrix} y_1' \\ y_2' \\ y_3' \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ a & b & c \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$
where a = ___, b = ___, c = ___ (FIB)

15. Given the equilibrium point of a 2 by 2 SDE is a star node.
Which of the following statements are true? (Pick all correct answers)

a. We can find infinitely many linearly independent eigenvectors for the underlying matrix.
b. All the solutions of the SDE have the form $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^{rt} \begin{pmatrix} a \\ b \end{pmatrix}$ where a and b can be any real numbers, and r is an eigenvalue of the underlying matrix.
c. The underlying matrix must be a scalar matrix.
d. All the trajectories on the phase portrait start near the origin and move away from the origin.
e. We need to find a generalised eigenvector to form a second solution for the fundamental set of solutions for the SDE.

Long Questions (4 questions; total 45 marks)

Q16 (10 marks)

Answer this question by writing in the exam answer booklet. Do not type in Examplify directly. Indicate Q16 at the top on each page of the answer to this question. Indicate the part numbers on the left margin. You may use MATLAB for this question. But you should show how your answers are derived.

The equation of a plane P in the 3D space is given by ax + by + cz = d for some unknown values a, b, c and d. Given that (1, 1, 3) and (2, -1, -2) are two points on the plane.

(i) [5 marks] Set up a linear system in a, b, c and d, and solve the system. Show your workings clearly. (You may use MATLAB to compute the standard matrix operations.)

(ii) [2 marks] Write down a basis for the solution space of the system in (i).

(iii) [3 marks] Write down an equation of the plane P if it contains the origin. Explain how you derive your answer.

Q17 (10 marks)

Answer this question by writing in the exam answer booklet. Do not type in Examplify directly. Indicate Q17 at the top on each page of the answer to this question. Indicate the part numbers on the left margin. You may use MATLAB for this question. But you should show how your answers are derived.

There are three quantities x, y, z which are subjected to the following constraints:

$$\begin{cases} x + 2y & = & 900 \\ y + z & = & 600 \\ x - 2z & = & 300 \end{cases}$$

(i) [2 marks] Show that the above system is inconsistent.

(ii) [5 marks] Find the least squares solutions of the above system. Show your workings clearly. (You may use MATLAB to compute the standard matrix operations.)

(iii) [3 marks] Suppose the values of x, y, z are all positive whole numbers. Find the particular least squares solution of the system that gives the smallest sum x + y + z. Explain how you derive your answer.

Q18 (12 marks)

Answer this question by writing in the exam answer booklet. Do not type in Examplify directly. Indicate Q18 at the top on each page of the answer to this question. Indicate the part numbers on the left margin. You may use MATLAB for this question. But you should show how your answers are derived.

In a certain city, there are three video streaming platforms A-flix, B-plus and C-prime. Among them there is a total of 63,000 subscribers (assuming each person only subscribe to one platform at any time). According to market research, for each month, there are movements among the subscribers of the three platforms: 4% and 1% of subscribers of A-flix switch to B-plus and C-prime respective; 6% and 4% of subscribers of B-plus switch to A-flix and C-prime respective; 4% and 4% of subscribers of C-prime switch to A-flix and B-plus respective.

(i) [3 marks] Let $\mathbf{v}_n = \begin{pmatrix} a_n \\ b_n \\ c_n \end{pmatrix}$, where a_n, b_n and c_n denote the number of subscribers of A-flix, B-plus and C-prime respectively in the nth month. Find the matrix $\mathbf{A}$ such that $\mathbf{v}_{n+1} = \mathbf{A}\mathbf{v}_n$.

(ii) [3 marks] Write down the n-th power $\mathbf{A}^n$ of $\mathbf{A}$ in the long run. Express the entries of the matrix in fractions. Show your workings clearly. (You may use MATLAB to compute the standard matrix operations.)

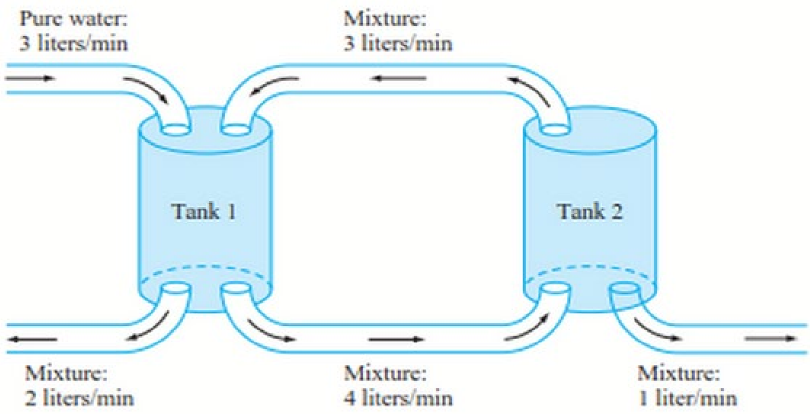
(iii) [3 marks] What should be the initial number of subscribers a_0, b_0 and c_0 of the three platforms so that a_n, b_n and c_n will remain unchanged every month? Explain how you derive your answer.

(iv) [3 marks] Is it possible that, with some initial number of subscribers a_0, b_0 and c_0, the three platforms will have equal market share in the long run? Justify your answer.

Q19 (13 marks)

Answer this question by writing in the exam answer booklet. Do not type in Examplify directly. Indicate Q19 at the top on each page of the answer to this question. Indicate the part numbers on the left margin. You may use MATLAB for this question. But you should show how your answers are derived.

Two tanks are connected by pipes as shown in the diagram. Tank 1 initially contains 20 litres of water in which 50 grams of chlorine are dissolved. Tank 2 initially contains 10 litres of water in which 50 grams of chlorine are dissolved. Beginning at time t = 0, pure water is pumped into Tank 1 at a rate of 3 litres per minute, while chlorine/water solutions are exchanged between the tanks and also flow out of both tanks at the rates shown.



(i) [3 marks] Set up an SDE in terms of the amount of chlorine $y_1(t)$ and $y_2(t)$ in Tank 1 and Tank 2 respectively at time t (in minutes)

(ii) [5 marks] Solve the initial value problem (IVP) with the SDE in (i) and the given initial conditions. Show your workings clearly.

(iii) [3 marks] Sketch the trajectory corresponding to the particular solution of the IVP in (ii).

(iv) [2 marks] Is it possible to adjust the initial amount of chlorine in the 2 tanks so that there will still be chlorine remains in the tanks in the long run? Justify your answers.